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## Paper 1: Application of Population-Based Meta-Heuristic Algorithms for Robust Initialization

### Bibliographic Information

- **Title:** Application of Population-Based Meta-Heuristic Algorithms for Robust Initialization in Mechanistic Pharmacometric Modeling
- **Authors:** Lee D, Lin C-W
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2026, vol 15, issue 8
- **DOI:** 10.1002/psp4.70301

**Paper Type:** METHODOLOGY

### Quick Take

This paper evaluates 19 population-based meta-heuristic algorithms (JSO, SOS, MA, etc.) as a pre-estimation strategy to generate robust initial THETA values for NONMEM-based mechanistic PK/PD modeling. On two ODE-based benchmark models, JSO, SOS, and MA consistently delivered near-optimal initial estimates with good computational efficiency, and the resulting NONMEM fits converged reliably even when some individual parameters (e.g.,  $Q$ ) were initially far off. The framework offers a data-driven, automated alternative to manual or literature-based initialization, though it requires user-specified parameter bounds and does not guarantee global optimality.

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### Executive Summary

The authors propose a systematic framework that uses population-based meta-heuristic algorithms to identify optimal or near-optimal initial parameter estimates (THETAs) for nonlinear mechanistic PK/PD models before likelihood-based estimation in NONMEM. Nineteen algorithms spanning evolutionary, swarm-based, and bio-inspired methods were benchmarked on a two-compartment PK model with Michaelis–Menten elimination and the Friberg myelosuppression model, using synthetic data generated from known true parameters. Performance was assessed via mean/best objective function values (OFVs), relative errors, and execution time under fixed population sizes, run-time limits, and iteration limits. Jellyfish Search Optimizer (JSO), Symbiotic Organisms Search (SOS), and Memetic Algorithm (MA) emerged as top performers across both models, balancing accuracy and computational cost. Increasing population size and iteration count improved OFV and parameter accuracy, and the algorithm-derived initial values supported successful NONMEM estimation, including in datasets with inter-individual variability. The framework is positioned as a practical pre-estimation strategy that reduces dependence on expert judgment and improves convergence stability, though it requires user-defined parameter bounds and does not guarantee convergence to the global minimum.

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## Scientific Context & Motivation

Nonlinear mixed-effects PK/PD modeling relies on iterative optimization of a nonconvex, multimodal objective function (typically  $-2 \log$ -likelihood). The choice of initial parameter values is critical: poor starting points can cause convergence failure, local minima, or implausible estimates. Traditional initialization relies on expert judgment, literature values, or exploratory data analysis, which becomes increasingly inadequate for complex mechanistic models with many parameters and strong nonlinearities (e.g., Michaelis–Menten elimination, feedback loops, transit compartments). Existing automated tools such as PsN’s retry mechanism are stochastic and non-directed, while pyDarwin and PharmPy focus on model-structure selection or workflow management rather than continuous parameter initialization. This paper addresses the gap by systematically evaluating population-based meta-heuristic algorithms as a global search pre-step to generate high-quality initial estimates for NONMEM.

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## ✂ Methodological Snapshot

The proposed framework uses population-based meta-heuristic algorithms as a pre-estimation step to generate initial THETA values for NONMEM. For a given mechanistic PK/PD model, the user specifies lower and upper bounds for each parameter. A population of candidate parameter vectors is initialized randomly within these bounds, and each candidate is evaluated by computing an objective function value (OFV, based on  $-2 \log$ -likelihood) from the observed data without invoking NONMEM. The population evolves over iterations using algorithm-specific update rules (e.g., jellyfish movement, symbiotic interaction, memetic local search), balancing global exploration and local refinement. After termination, the candidate with the lowest OFV is selected as the initial estimate for subsequent NONMEM estimation. The framework was evaluated on two ODE-based models: a two-compartment PK model with linear and Michaelis–Menten elimination, and the Friberg myelosuppression model. Nineteen algorithms were compared under identical population sizes and run-time/iteration limits, with 15 independent runs each. Top performers (JSO, SOS, MA) were further tested under varying population sizes and iteration counts, and robustness was assessed using datasets with inter-individual variability.

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## ☒ Statistical Framework

The statistical framework is based on the assumption that the objective function (OFV, typically  $-2 \log$ -likelihood) of a nonlinear PK/PD model is nonconvex and multimodal, making gradient-based local optimization sensitive to starting values. Population-based meta-heuristic algorithms perform stochastic global search without requiring gradient information or convexity assumptions. The framework assumes that candidate parameter sets with lower pre-estimation OFVs are more likely to lie in regions of the parameter space that support successful convergence during subsequent likelihood-based estimation (e.g., NONMEM’s FO or FOCE). The OFV used during metaheuristic search is computed directly from the simulated data (pre-NONMEM), without incorporating random effects (IIV) in the primary evaluation. The theoretical basis is that population diversity and iterative refinement enable exploration of multiple regions of the parameter space, reducing the risk of premature convergence to poor local minima. The approach

does not guarantee global optimality but aims to provide ‘good enough’ starting points for local refinement.

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### **Estimator Behavior**

The paper evaluates the behavior of the metaheuristic-derived initial estimates in terms of their ability to support subsequent NONMEM estimation. Key observations: (1) JSO, SOS, and MA consistently produced initial estimates that led to successful NONMEM convergence with final estimates closely aligned with true values; (2) even when individual parameters (e.g.,  $Q$  in the PK model) deviated >10-fold from true values, NONMEM still converged to accurate estimates, indicating that the initial estimates placed the optimizer in a favorable region of the parameter space; (3) increasing population size and iteration count improved the accuracy of initial estimates (lower OFV, smaller deviations from true values), though with diminishing returns and increased computational cost; (4) the framework showed robustness in the presence of IIV, with final NONMEM estimates centered near true values across 10 simulated datasets. The paper does not report formal bias/variance metrics for the initial estimates themselves, but the convergence success and accuracy of final NONMEM estimates serve as indirect evidence of estimator behavior.

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### **Validation Design**

The validation design uses synthetic datasets generated from known true parameter values for two benchmark models: (1) a two-compartment PK model with linear and Michaelis–Menten elimination, and (2) the Friberg myelosuppression model. Each model produced concentration–time profiles for nine subjects at three dose levels. Nineteen metaheuristic algorithms were evaluated with identical population sizes (100 for PK, 80 for PK/PD) and run-time limits (1 h for PK, 5 h for PK/PD) or iteration limits (100 for PK, 80 for PK/PD), with 15 independent runs per algorithm. Performance was assessed via mean and best OFVs, relative errors relative to the OFV from true parameters, and mean execution time. Top-performing algorithms (JSO, MA, SOS) were further evaluated under varying population sizes and iteration counts. Robustness was assessed using 10 simulated datasets with inter-individual variability (50% CV for PK model) for each model, with algorithm-derived initial estimates used for NONMEM FOCE estimation. Final NONMEM estimates were compared to true values to assess accuracy and precision.

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### **Applicability Boundaries**

The framework is applicable to nonlinear mechanistic PK/PD models where manual initialization is challenging due to nonconvex, multimodal parameter spaces. It works best when: (1) the model has many parameters and strong nonlinearities (e.g., Michaelis–Menten elimination, feedback loops, transit compartments); (2) the user can specify physiologically plausible parameter bounds; (3) computational resources allow for population-based search (minutes to hours depending on model complexity). The framework may be less useful for simple models where traditional initialization works well, or when parameter bounds cannot be reasonably defined. Limitations: (1) the approach does not guarantee global optimality; (2) the pre-NONMEM OFV may not perfectly align with the NONMEM objective function, especially when random effects are present; (3)

performance may degrade with model misspecification or sparse data; (4) the user must review results for physiological plausibility; (5) the framework requires additional computational time compared to manual initialization, though this is offset by reduced convergence failures.

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### **Comparison to Alternatives**

Compared to traditional manual initialization (expert judgment, literature values, exploratory data analysis), the metaheuristic framework offers a data-driven, automated, and reproducible approach that can explore a wider parameter space and reduce dependence on user expertise. Compared to PsN's retry mechanism, which performs stochastic random resets within  $\pm 10\%$  of initial values without feedback, population-based algorithms use adaptive search that learns from previous iterations, potentially finding better starting points. Compared to pyDarwin (which uses GA/PSO for model-structure selection) and PharmPy (which provides workflow tools), the proposed framework focuses specifically on continuous parameter initialization. Compared to gradient-based multi-start methods, population-based methods do not require derivatives and can handle nonconvex surfaces more robustly, though they may be less efficient for smooth, well-behaved problems. The paper does not provide a direct head-to-head comparison with these alternatives, but positions the framework as a complement to existing tools.

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### **Implementation Guidance**

Implementation steps: (1) Define the mechanistic PK/PD model as an ODE system and specify lower/upper bounds for each parameter based on prior knowledge, literature, or physiological plausibility; (2) Implement the metaheuristic algorithm in Python using the pyMetaheuristic library (version 1.9.5) or similar; (3) Compute OFV for each candidate parameter set directly from the data (without invoking NONMEM) using numpy/scipy; (4) Run the algorithm with a population size  $>50$  and  $\geq 50$  iterations as a starting configuration (larger settings for more complex models); (5) Select the candidate with the lowest OFV as the initial THETA values; (6) Use these as initial estimates in NONMEM (FO or FOCE) and verify convergence and physiological plausibility. Computational cost: for the PK model, satisfactory performance (RE  $<1\%$ ) required  $\sim 8\text{--}70$  min; for the Friberg model, RE  $<0.8$  required  $\geq 90$  min. Multi-core execution (e.g., 24 cores) is recommended to parallelize fitness evaluations. The paper used Python 3.10.16 with numpy 2.2.6, scipy 1.15.3, pyMetaheuristic 1.9.5, and joblib 1.5.1 on a Windows 11 system with an Intel i9-13950HX processor and 128 GB RAM.

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### **☒ Key Findings**

- 1) Among 19 algorithms, JSO, SOS, and MA consistently achieved the lowest mean and best OFVs on both benchmark models, with JSO showing particularly favorable accuracy-to-computation-time trade-offs.
- 2) Increasing population size and iteration count improved OFV and parameter accuracy, with JSO reaching relative error  $<1\%$  on the PK model with population size  $>50$  and  $\geq 50$  iterations ( $8\text{--}70$  min), and relative error  $<0.8$  on the Friberg model with similar settings ( $\geq 90$  min).
- 3) Algorithm-derived initial estimates supported successful NONMEM estimation; even when JSO's estimate of Q deviated  $>10$ -fold from the true value,

NONMEM converged to estimates closely aligned with true values. 4) In robustness assessments with inter-individual variability (10 datasets per model), JSO-, MA-, and SOS-derived initial THETAs supported successful NONMEM FOCE estimation with final estimates centered near true values. 5) Performance was not clearly determined by algorithm category (evolutionary vs. swarm vs. bio-inspired), suggesting that specific search mechanisms and problem characteristics matter more than broad taxonomy.

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## **Strengths & Limitations**

### **Strengths**

- Systematic evaluation of 19 diverse population-based meta-heuristic algorithms on two representative mechanistic ODE-based PK/PD models
- Clear performance metrics (mean/best OFV, relative error, execution time) with multiple independent runs (15 per algorithm)
- Demonstrates practical value by using algorithm-derived initial estimates in subsequent NONMEM estimation, including robustness checks with inter-individual variability
- Provides practical guidance on population size and iteration settings (e.g., population >50,  $\geq 50$  iterations) for achieving satisfactory performance
- Addresses a real bottleneck in pharmacometric workflows (initial value selection) and offers an automated, data-driven alternative
- Computational implementation is transparent (Python, pyMetaheuristic, multi-core) and reproducible

### **Limitations (Acknowledged by Authors)**

- The framework requires user-defined parameter bounds; setting appropriate bounds relies on prior knowledge or exploratory analysis
- The approach does not guarantee convergence to the global minimum; local minima with favorable OFVs may still exist
- The primary evaluation used nonlinear models without random effects; extension to full nonlinear mixed-effects models requires incorporating IIV parameters
- Analysts must review results to ensure parameter estimates remain physiologically plausible

### **Limitations (Expert Review)**

- The benchmark models were fitted with the same ODE structure used for data generation, which may overstate performance relative to real-world model misspecification
- The OFV used during metaheuristic search is a pre-NONMEM objective function (likely a simple least-squares or likelihood without random effects), which may not perfectly align with the NONMEM objective function; the correlation between pre-NONMEM and NONMEM OFVs is not formally characterized
- Only two benchmark models were used; generalizability to other model structures (e.g., with absorption delays, multiple dosing, or complex covariate effects) is untested
- The choice of ‘true’ parameter values and bounds may influence algorithm rankings; sensitivity of rankings to bound specification is not explored

- The robustness assessment with IIV used only 10 datasets per model and focused on the top three algorithms; statistical power to detect differences is limited
- No formal statistical comparison (e.g., hypothesis tests) between algorithms; rankings are based on descriptive summaries
- The paper does not report the exact objective function used during metaheuristic search (e.g., whether it included residual error modeling), which limits reproducibility of the OFV values

### Generalizability

The framework is conceptually generalizable to any nonlinear mechanistic PK/PD model, but the empirical findings (algorithm rankings, recommended settings) are based on two specific ODE models with synthetic data generated under the same model structure. Performance may vary with model complexity, data sparsity, and the presence of random effects. The approach is most likely to benefit complex models with many parameters and strong nonlinearities where manual initialization is challenging; for simple models, traditional initialization may suffice.

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### Key Equations

#### Objective Function Value (OFV) for metaheuristic search

$$\text{OFV} = -2 \log L(\theta; y)$$

The objective function minimized by the metaheuristic algorithms during the pre-estimation search; lower OFV indicates better fit to the observed data.

#### Relative Error (RE) of OFV

$$\text{RE} = \frac{\text{OFV}_{\text{alg}} - \text{OFV}_{\text{true}}}{\text{OFV}_{\text{true}}}$$

Used to quantify the deviation of an algorithm's best or mean OFV from the OFV obtained with the true parameter values.

#### Two-compartment PK model with Michaelis–Menten elimination (structural form)

$$\frac{dA_1}{dt} = -\left(\frac{V_{\max}}{K_m + C_1} + \frac{CL}{V_1}\right)C_1 + \frac{Q}{V_2}C_2, \quad \frac{dA_2}{dt} = \frac{Q}{V_1}C_1 - \frac{Q}{V_2}C_2$$

The ODE system describing drug amounts in central (A1) and peripheral (A2) compartments, with saturable (Michaelis–Menten) and linear elimination.

#### Friberg myelosuppression model (proliferation feedback)

$$\frac{dP}{dt} = k_{\text{prol}} \cdot P \cdot \left(\frac{C_0}{C}\right)^\gamma \cdot (1 - E_{\text{drug}}(C)) - k_{\text{tr}} \cdot P$$

The proliferation compartment in the Friberg model, where the feedback term  $(C_0/C)^\gamma$  regulates proliferation based on circulating neutrophil counts, and drug exposure inhibits proliferation.

## Figures & Tables

- **Figure 1:** Trends in OFVs over iterations for representative population-based optimization algorithms (top five by average OFV) for the 2CMT PK model (a) and Friberg myelosuppression model (b).
    - *Significance:* Illustrates the convergence behavior of the best-performing algorithms and demonstrates that population-based search effectively reduces OFV over iterations, avoiding local minima.
  - **Figure 2:** Execution times (left panels) and objective function values (right panels) of the JSO algorithm under varying population sizes and iteration numbers for the PK model (a,b) and Friberg model (c,d).
    - *Significance:* Provides practical guidance on the trade-off between computational cost and optimization accuracy, showing that larger populations and more iterations improve performance but increase runtime.
  - **Figure 3:** Accuracy and precision of PK parameter estimates using JSO, MA, and SOS across 10 simulated 2CMT PK datasets with inter-individual variability.
    - *Significance:* Demonstrates the robustness of the proposed framework under conditions closer to real nonlinear mixed-effects modeling, showing that algorithm-derived initial estimates support successful NONMEM estimation.
  - **Figure 4:** Accuracy and precision of PD parameter estimates using JSO, MA, and SOS across 10 simulated Friberg myelosuppression model datasets with inter-individual variability.
    - *Significance:* Confirms that the framework remains effective for a more complex mechanistic model with feedback and transit compartments, even in the presence of IIV.
  - **Table 1:** Performance of 19 algorithms on the 2CMT PK model with MM elimination, showing mean/best OFV and relative errors under 1-hour runtime and 100-iteration limits.
    - *Significance:* Provides the primary benchmark comparison, identifying JSO, MA, SOS, TLBO, and IWOA as top performers and highlighting poor convergence for RS, SPBO, BA, and HSA.
  - **Table 2:** Performance of 19 algorithms on the Friberg myelosuppression model, showing mean/best OFV and relative errors under 5-hour runtime and 80-iteration limits.
    - *Significance:* Extends the benchmark to a more complex model, confirming SOS, JSO, GA, MA, and MBO as strong performers and demonstrating the framework’s applicability to PD models.
  - **Table 3:** Parameter bounds, JSO-derived initial estimates, true values, NONMEM estimates, and RSEs for both benchmark models.
    - *Significance:* Shows that JSO-derived initial estimates, even when deviating substantially from true values (e.g.,  $Q$ ), still support successful NONMEM convergence to estimates closely aligned with true values.
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## Code & Reproducibility Assessment

The paper describes the computational environment (Python 3.10.16, numpy 2.2.6, scipy 1.15.3, pyMetaheuristic 1.9.5, joblib 1.5.1, multi-core execution on Windows 11) but does not provide a link to publicly available code or scripts. The use of the pyMetaheuristic library suggests the algorithms are implemented in a standard package, but the specific configuration (e.g., algorithm

hyperparameters, OFV computation details) is not fully specified. Reproducibility would require contacting the authors or reconstructing the framework from the methods description.

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### **Supplementary Materials**

Supplementary materials include Figure S1 (model schematics), Figure S2 (general metaheuristic workflow), Table S1 (algorithm categorization), Table S2 (parameter bounds), Figure S4 (simulated data), Figure S5 (OFV paths for all algorithms on PK model), Figure S6 (JSO parameter estimates vs. configuration for PK model), Figure S7 (OFV paths for all algorithms on Friberg model), and Figure S8 (JSO parameter estimates vs. configuration for Friberg model). These provide additional detail on algorithm taxonomy, convergence trajectories, and sensitivity of parameter estimates to optimization settings.

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### **Future Directions**

- 1) Extend the framework to full nonlinear mixed-effects models by optimizing both fixed effects and variance components (IIV, residual error) simultaneously.
  - 2) Evaluate the approach on a broader range of model structures (e.g., with absorption delays, enterohepatic recirculation, target-mediated drug disposition) and real clinical datasets.
  - 3) Investigate the relationship between pre-NONMEM OFV and final NONMEM OFV to establish formal criteria for accepting algorithm-derived initial estimates.
  - 4) Develop adaptive strategies for setting parameter bounds based on prior distributions or physiological constraints.
  - 5) Benchmark against other initialization strategies (e.g., PsN retry, multi-start gradient methods) in a head-to-head comparison.
  - 6) Explore the use of metaheuristic-derived parameter distributions to inform identifiability analysis and model reduction.
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### **Expert Commentary**

This paper addresses a practical and often underestimated challenge in pharmacometrics: the sensitivity of nonlinear mixed-effects estimation to initial values. The systematic comparison of 19 metaheuristic algorithms is valuable, and the identification of JSO, SOS, and MA as top performers provides actionable guidance. However, the study’s reliance on synthetic data generated under the same model structure used for fitting is a notable limitation; real-world model misspecification and data sparsity may reduce the apparent advantage of these algorithms. The framework’s dependence on user-specified bounds is also a double-edged sword—it provides flexibility but requires domain knowledge. From a statistical perspective, the pre-NONMEM OFV used during metaheuristic search is not the same as the NONMEM objective function (which includes random effects), so the optimality of the initial estimates is not guaranteed to translate directly to the final estimation. Nevertheless, the empirical demonstration that NONMEM converges successfully from these starting points is compelling. I would recommend this approach as a complement to, not a replacement for, careful exploratory analysis and model diagnostics. The paper would be strengthened by formal statistical comparisons between algorithms and by testing on real datasets with known convergence challenges.

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### **Bottom Line**

For pharmacometricians facing convergence failures or unreliable fits in complex mechanistic PK/PD models, population-based meta-heuristic algorithms—particularly Jellyfish Search Optimizer, Symbiotic Organisms Search, and Memetic Algorithm—offer a practical, automated pre-estimation strategy to generate robust initial THETA values. The framework requires only user-specified parameter bounds and can be implemented in Python before NONMEM estimation. Recommended starting configuration: population size >50 and ≥50 iterations, with larger settings for more complex models. This approach reduces dependence on expert judgment, improves convergence stability, and supports parameter identifiability assessment, though it does not replace the need for careful model diagnostics and physiological plausibility review.

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### **☒ Figures**

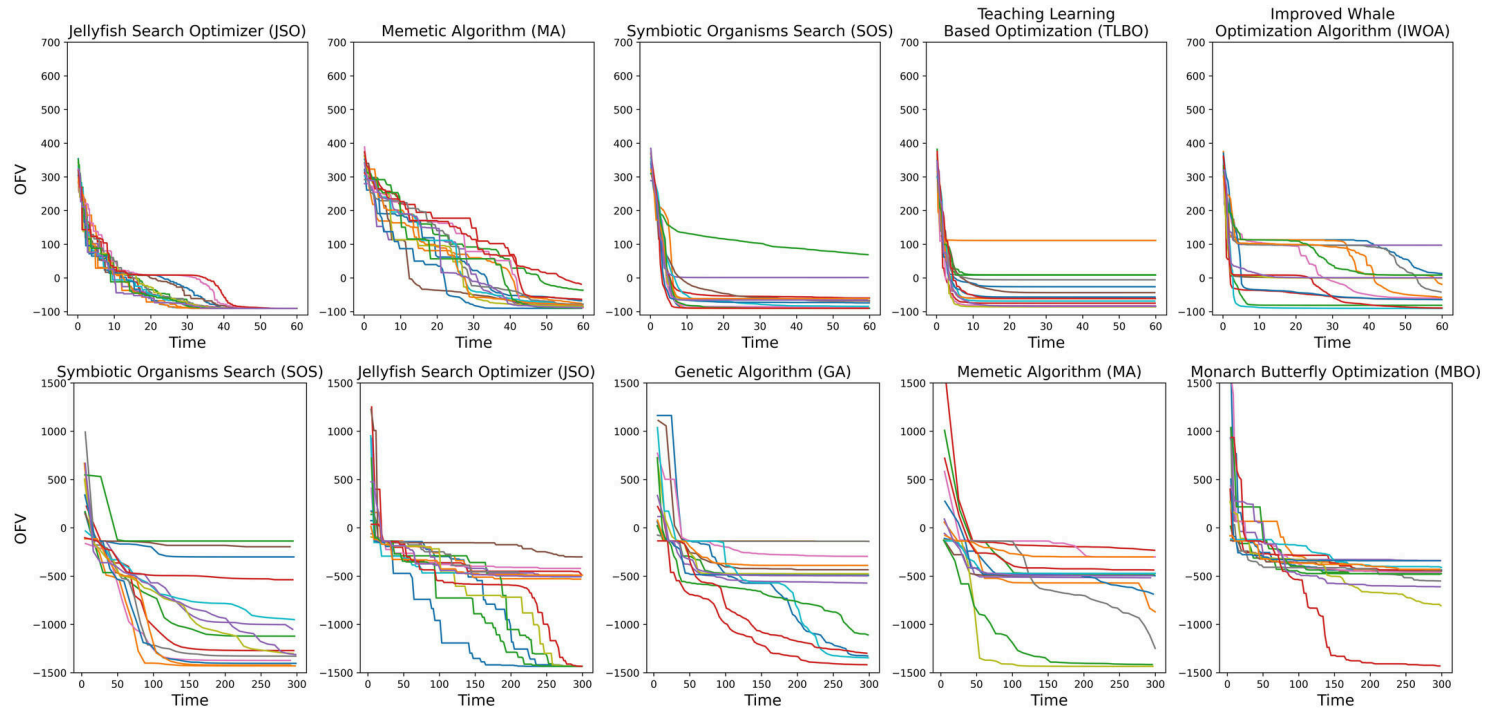


Figure 1: Trends in OFVs over iterations for representative population-based optimization algorithms. OFVs over time for the top five algorithms based on average OFV in th

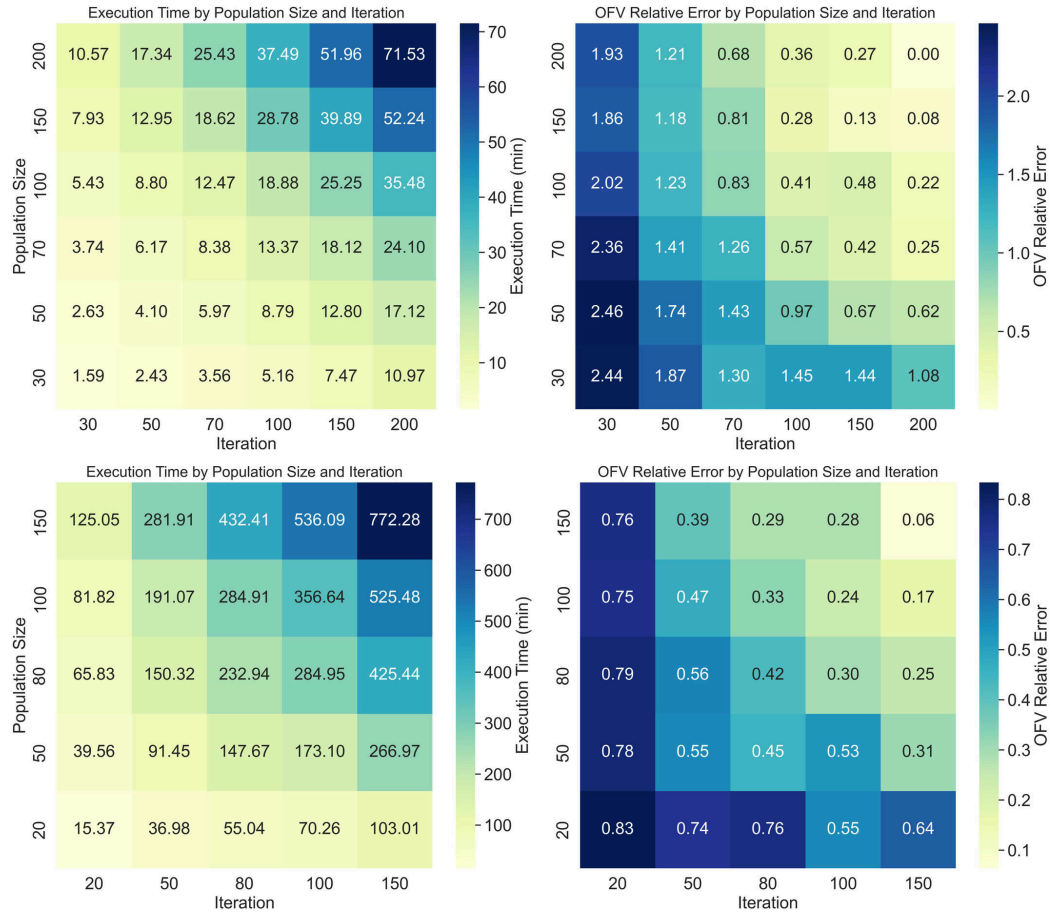


Figure 2: Execution times and OFVs of the JSO algorithm under varying population sizes and iterations. Execution times (left panels) and objective function values (right p

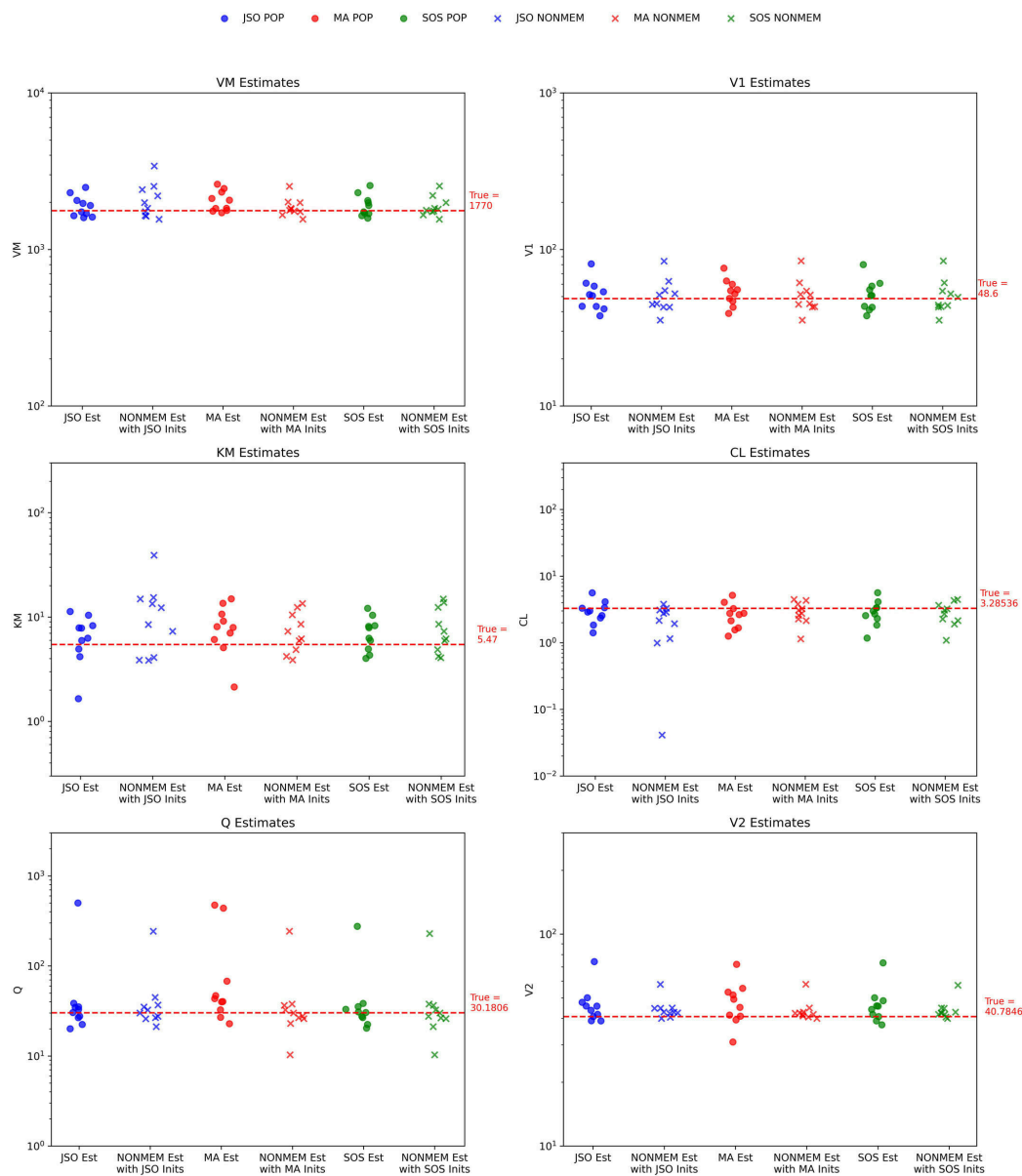


Figure 3: Accuracy and precision of PK parameter estimates using JSO, MA, and SOS across 10 simulated 2CMT PK datasets with inter-individual variability. The proposed meth

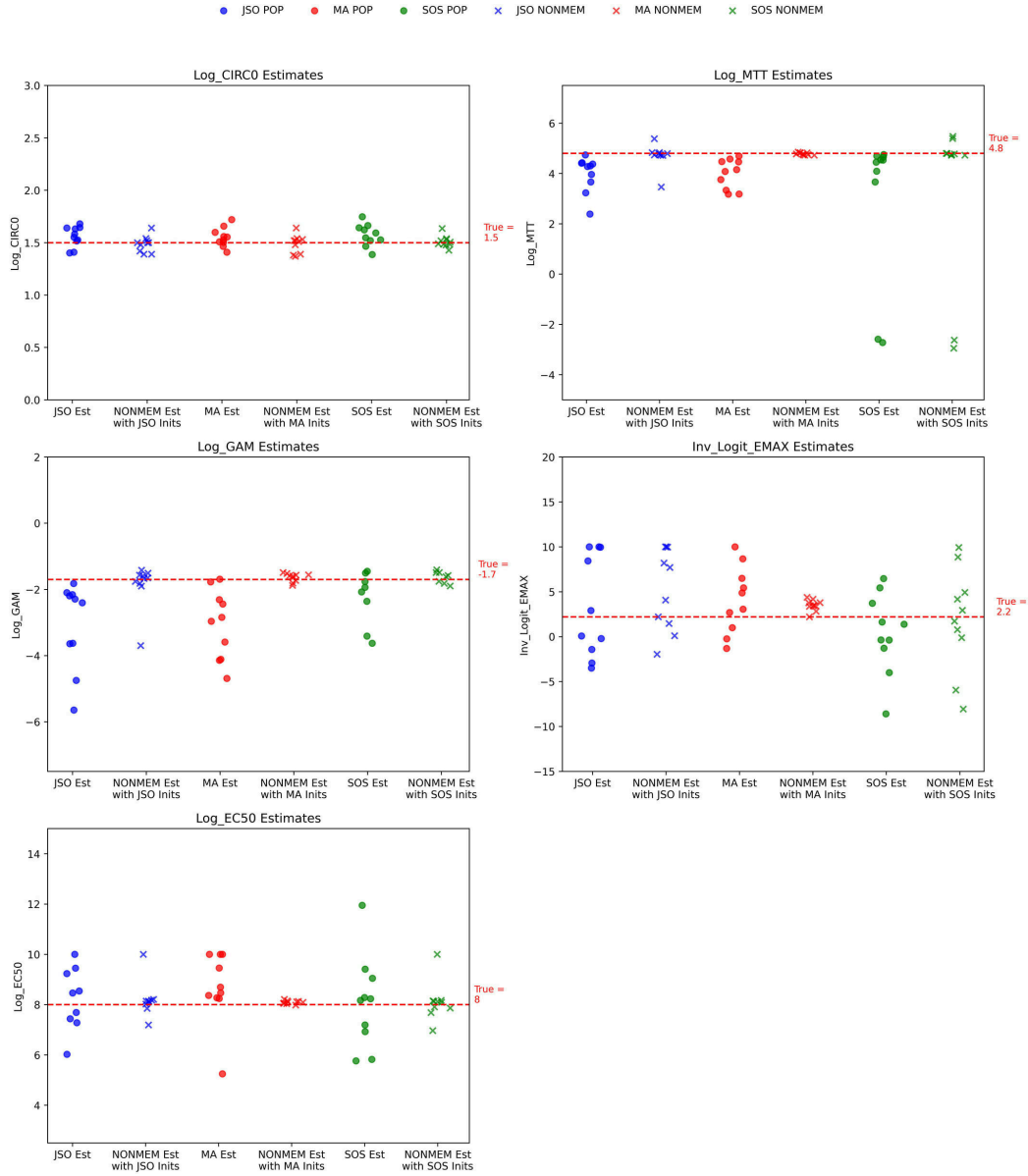


Figure 4: Accuracy and precision of PD parameter estimates using JSO, MA, and SOS across 10 simulated Friberg Myelosuppression model datasets with inter-individual variability